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RESEARCH ARTICLE

Nanoplastics-induced oxidative stress, antioxidant defense, and physiological response in exposed Wistar albino rats

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Abstract

Nowadays, plastic pollution and in particular nano(micro)plastics is considered as an issue of global concern in environmental samples. The present work was conducted to clarify the oxidative stress of polystyrene nanoplastics (PS-NPs) exposure and physiological response of male Wistar rats. Animals were treated orally with PS-NPs at four doses (1, 3, 6, and10 mg/kg-day) for 5 weeks. Results demonstrated the accumulation of PS-NPs through whole body scanning and also a dose-dependent increase in the production of reactive oxygen species (ROS). Alterations in antioxidant responses including serum levels of catalase (CAT) and total glutathione content were noticed, but not superoxide dismutase (SOD), pointing towards the perturbation of redox state induced by exposure conditions. Biochemical parameters viz. glucose, cortisol, lipase, lactate, lactate dehydrogenase (LDH), alkaline phosphatase, gamma-glutamyl transpeptidase (GGT), triglycerides, and urea showed a significant increase, while total protein, albumin, and globulin levels showed an appreciable decline. The pattern of associations noticed with AChE activity and biochemical responses in our study suggests the possibility that a neurobehavioral effect or dysfunctions in energy metabolism may be the potential modes of action, possibly through stress response as well as liver function. Perturbations of creatinine and uric acid levels are indeed plausible biological explanations for the association with kidney dysfunction. Although we provided a new scientific clue for exploring the biological consequences of NPs which might induce effects such as oxidative stress relating to the induction of antioxidant enzymes, the results warrant additional research with a larger sample size.

Keywords Polystyrene nanoplastics · Rat · Bioaccumulation · Serum biomarkers · Oxidative stress

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Introduction

Reports of plastic pollution have mounted serious environmental concerns for the potential adverse influences on marine, wildlife, and public health. Even with concerted global effort, the amount of particulate plastics—small plastic fragments or beads (i.e., 5 mm down to the nanometer range)—in the environment is continuing to grow (Frias and Nash 2019). Although plastic particles have been highlighted to induce various physiological disorders, the cellular and molecular mechanism-linking exposure to microscopic plastic particulates and potential health hazards remains to be elucidated. Nonetheless, oxidative stress, genotoxicity, neurotoxicity, inflammation as well as dysregulation of lipid and energy metabolism have been the proposed mechanisms, among others, reported in literature (Lu et al. 2016; Prokić et al. 2019).

The principles of plastic particulates toxicity have been extensively reviewed (Wong et al. 2020; Yong et al. 2020), even though evidences of harmful effects have mostly restricted to observations on marine and aquatic species and howbeit microplastics (MPs). Public health and environmental concerns, however, increasingly drive efforts to unravel the potential effects following exposure to nanoplastics (NPs—fragments ≤ 100 nm) in humans which could be a future prime concern. Compared with larger plastic particulates, NPs are of particular concern, accounting for their plausible interaction with living organisms as they are more likely to cross biological barriers, accumulate in organs and affect the function of cells causing systemic exposure (Amereh et al. 2019). This becomes even more complicated as the contamination of nano (micro) plastics on terrestrial environment has been argued to be 4 to 23-fold greater than in the marine (Horton et al. 2017), and more remarkably, mammals at high trophic level are regularly in the throes of exposure to microscopic plastic debris through drinking water, inhalation, and food chain all over the world (Rist et al. 2018).

Blood biochemistry deals with the processes that occur within and between living cells which in turn relates greatly to the prediction of tissues, organs, and organism structure and function (Banaee et al. 2014). By the way, oxidative stress reflects a disturbed balance between the systemic generation of potentially damaging reactive oxygen and nitrogen species (ROS and RNS, respectively) and an organism capacity to readily detoxify the reactive intermediates through antioxidant defenses or to repair the resulting damage. Chemically reactive compounds containing oxygen including peroxides, superoxide, and hydroxyl radicals as well as singlet oxygen represent ROS (Hayyan et al. 2016). ROS has become an area of great concern for scientists due to the fact that this pathway of programmed health deterioration has also resulted in dysbiosis (Sharma and Padwad 2019), poor fertilization and embryonic development (Zhuang et al. 2020), pregnancy loss (Almada et al. 2020), birth defects (including autism) (Lee et al. 2019), and childhood cancer (Agarwal et al. 2014). Oxidative stress and biochemical processes are closely related to several biotic and abiotic factors; among them, environmental insults are considered key players (Chen et al. 2020). The literature already includes several reports of adverse effects of different pollutants on oxidative stress and metabolic profiles of aquatic and terrestrial organisms; however, the effects of NPs have been poorly elucidated (Chen et al. 2020; Kim et al. 2019; Yong et al. 2020). Even though MPs (either virgin or with sorbed chemical pollutants) are known to induce various apical endpoints, there is a paucity of toxicological investigations concerned with cellular biomarkers to test metabolic defects of nanosized plastics, particularly in rodent species usually used for toxicity studies on which to base risk assessment for humans. Nevertheless, the literature on this topic has only just started to increase (Yang et al. 2019; Lu et al. 2018; Zheng et al. 2021).

There are innate antioxidant defenses to cope with the threat in the body. Rampaging free radicals are neutralized by the action of a battery of enzymatic (e.g., superoxide dismutase, CAT), glutathione peroxidase (GPX), glutathione S-transferase (GSH), and glutathione reductase (GR), among others) and nonenzymatic (e.g., glutathione, ubiquinol, uric acid) components during normal metabolism in the body where delay or inhibit cellular damage through their free radical scavenging property (Nita and Grzybowski 2016). Acetylcholine (Ach)—cholinergic system's neurotransmitter—on the other hand, takes a leadership role in nervous system and assessing the effects of environmental insults (Yen et al. 2011). It was ever reported that MPs could pose neurotoxicity through the inhibition of AChE activity (Deng et al. 2017; Oliveira et al. 2013). Some studies have also addressed the relationship between toxicity mechanisms and neurobehavioral alterations in mice exposed to pristine polystyrene MPs (Deng et al. 2017; Yong et al. 2020); nevertheless, knowledge about the serious and emerging problem of the toxic effects of plastic detritus and particularly NPs in rodent species addressing the risk for humans is presently lacking.

On the premise of this point of view, and considering an unappreciated set of responses at molecular signaling and stress response pathways in an in vivo model, we suggest them as early warning signals for predicting adverse effects at higher biological organization levels (population or community) as a consequence of exposure to NPs. This response was assessed based on the measurements of oxidative stress biomarkers through either free radical production or key antioxidant enzymes activities, as well as some biochemical parameters related to the liver and kidney function. The release of lactate dehydrogenase (LDH) was also worked out to figure the integrity of cells induced by PS-NPs. Adult male Wistar rats were used as a model species to investigate the implications on human health and body, because they are known as the chief model in numerous fields, such as neurobehavioral and toxicological studies (Sengupta 2013). The work was conducted to clarify the stress response following the relatively long-term (35 days) exposure with PS-NPs and to elucidate the underlying mechanisms. The results obtained provide detailed information on the potential health risks of microscopic plastic particulates exposure to mammals including human.

Materials and methods

Nanoplastics, chemicals, and reagents

Pristine PS-NPs with nominal diameters of 25 and 50 nm were delivered by a private plastic company (G. Kisker GbR, Steinfurt, Germany). Binary mixtures of nanoplastic suspensions were freshly prepared every day, just before treatment, by adding predetermined volumes of 25 and 50 nm PS-NPs stocks to distilled water (pH 6.9 ± 0.1) considering equal weights of each chemical. In parallel with toxicological tests, fluorescent PS particles with respective excitation and emission wavelengths of 535 and 550 nm were used to assess the accumulation and distribution of particulates in the body. PS spheres showed 38.92 nm mean hydrodynamic diameter with negative surface charges (−9.27 ± 3.57 mV), which were determined through laser diffraction and time-resolved dynamic light scattering (DLS) measurements (Malvern Instruments Ltd, Malvern, UK) (Fig. 1).

All chemicals and reagents including reduced nicotinamide adenine dinucleotide (NADH), nitroblue tetrazolium (NBT), phenazin methosulphate (PMS), ammonium molybdate, hydrogen peroxide (H_2O_2), 5,5′-dithiobis-(2-nitrobenzoic acid) (DTNB), and acetylthiocholine were purchased from Sigma Chemical (St. Louis, MO, USA). Kit for analyzing ROS was purchased from MyBioSource, USA.

Animals and exposure experiments

Thirty young sexually mature male rats weighing 190 ±10 g were kept under standard laboratory conditions (i.e., room temperature 22 ± 2°C, 50–60% relative humidity, 12 h dark–light cycle with 12–14 air changes per hour). Commercial food and water were available ad libitum. All experimental processes were in strict accordance with the treatment and care of laboratory animals published by the US National Institute of Health guidelines for the care and use of laboratory animals and received approval from the Research, Ethical Committee of the Faculty of Public Health and Safety, Shahid Beheshti University of Medical Science, Iran. After 5 days of acclimatization, rats were randomized into four experimental and control groups, with 6 rats in each treatment group and administrated with 1, 3, 6, and 10 mg/kg-day. The exposure doses used here was based on the Precautionary Principle to reveal the target mechanisms first before evaluating the risk at environmentally relevant concentrations. NPs were prepared by treating the stock suspensions in an ultrasonic bath for 15 min at 42 W/L and administered in 500 μL Milipore Mili-Q water via oral gavage for 35 consecutive days. The control group received the same volume of distilled water only. At the end of exposure trials, animals were euthanized by chloroform solution. Trunk blood was collected and serum isolated via centrifugation (6000 rpm for 5 min at 4°C) in vacutainer tubes, transferred to 1.5 mL microcentrifuge tubes and stored at −80°C before analysis.

Serum biochemical analysis

Perturbations in biological responses at the serum level were determined, which encompassed a suite of neurotoxic responses in terms of acetyl cholinesterase activity, oxidative stress-related biomarkers, including ROS profiling and glutathione content, as well as enzymatic activities of antioxidant defense system including superoxide dismutase and catalase. Biochemical parameters (including total protein (TP), albumin, glucose,

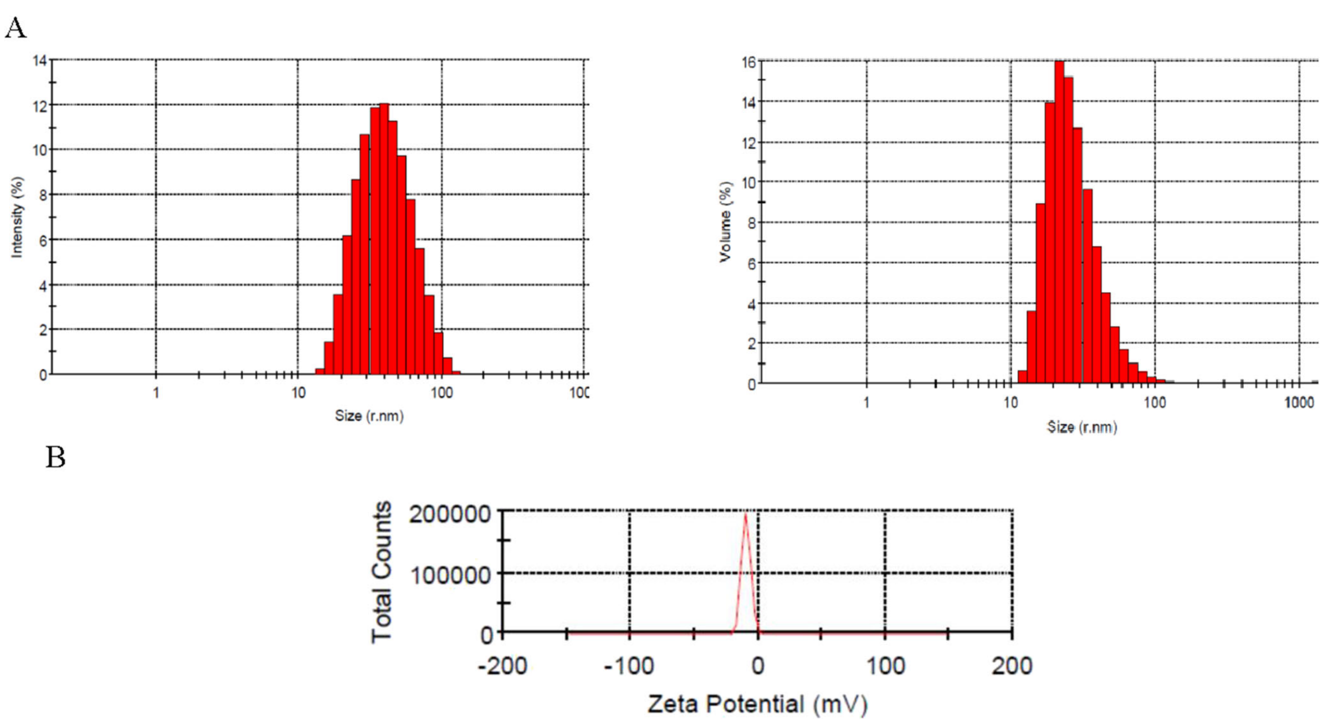


Fig. 1 Characterization of PS NPs. **A** The hydrodynamic diameter **B** and zeta potential

lactate, lactate dehydrogenase (LDH), alkaline phosphatase, gamma-glutamyl transpeptidase (GGT), lipase, triglycerides, and urea) were indeed worked out that could be associated with the exposure to PS-NPs.

Acetyl cholinesterase activity was measured by the modified Ellman's colorimetric method (Ellman et al. 1961), assessed by the addition of Ellman's reagent—DTNB—using AChEI (acetylcholine) as substrate, for the estimation of respective thiocholine. The absorbance was measured at 405 nm in ELISA plate reader (Biotek ELx 800, USA), to estimate the amount of ChE liberated by a reaction proportional to the AChE activity.

ROS determination was carried out using a 96-well-plate ELISA Kit according to the manufacturer's instructions and detected at 450 nm in ELISA reader. Serum GSH levels were determined using the dithiobis nitrobenzoic acid (DTNB) method—Ellman's method. Accordingly, 60 μL of sample was used for GSH assay by Ellman's reagent (19.8 mg of DTNB in 100 mL 0.1% sodium nitrate and phosphate buffer (pH 7.4)). The absorbance was measured at 412 nm (Ellman 1959). The method is based on a colorimetric quantification that can be measured by the formation of GS-TNB complex from DTNB (5-5′-dithiobis [2-nitrobenzoic acid]) in which a yellow color develops because of the DTNB reduction. The glutathione concentration was determined through spectrophotometric method by absorbance at 415 nm.

The assay for superoxide dismutase activity involves the inhibition of nitro blue tetrazolium reduction. The formation of amino blue tetrazolium formazan is in the nicotinamide adenine dinucleotide, phenazine methosulphate, and nitro blue tetrazolium (NADH–PMS–NBT), according to the method of Kakkar et al. (1984), which is measured spectrophotometrically at 560 nm. One unit of enzyme activity is defined as that amount of enzyme that results in 50% inhibition of NBT reduction/mL serum.

Catalase activity was determined as previously described (Goth 1991), based on the formation of a stable complex of hydrogen peroxide with ammonium molybdate. Approximately 20 μL of sample was incubated in 1 mL reaction mixture containing 65 mM hydrogen peroxide in 60 mM sodium-potassium phosphate buffer (pH 7.4) at 25 °C for 4 min. The enzymatic reaction was stopped with 1 mL of 32.4 mM ammonium molybdate and the concentration of the developed yellow stable complex of molybdate and hydrogen peroxide was measured at 405 nm.

Colorimetric measurements of the selected biochemical parameters were performed in triplicate on an automatic dry-chemistry analyzer system (TS/technology Alpha—Classic). Globulin concentrations were determined by deducting TP from the albumin concentration.

Cortisol was quantified in the serum through ELISA using commercially available kit (Catalog no. DCM001-12) provided from DiaMetra, Italy, following the manufacturer's protocol.

Tissue accumulation and fluorescence imaging

Fluorescently labeled PS-NPs were dispersed in 500 μL Milipore Mili-Q water and treated by ultrasonic vibration. The mixed solution was then given once daily (10 mg/kg-day) by oral gavage. The whole body was scanned for fluorescence with a fluorescence imager (KODAK InVivo Imaging System FX Pro, USA) at excitation/emission wavelengths of 535/550 nm.

Data analysis and statistics

All raw data were presented as means and standard deviation (±SD). The data were initially submitted for normality (Kolmogorov–Smirnov test) and homoscedasticity (Levene's test). Analysis of variance (ANOVA) was employed to assess the significant differences among treatments where was set a priori at the $p \leq 0.05$ for all statistical analyses. The analysis was followed by Tukey's multiple range test to compare individual treatments. A variance explained statistic eta-squared (n^2) was used as the reported effect size (ES), and though ascertaining what constitutes a small/medium/large effect is primarily a function of the context/discipline. For this report, 0.01/0.059/0.138 was small/medium/large (Cohen 1988). The strength of relationships between lipids and proteins were assessed using Pearson correlations. Statistical analyses were performed using STATA (version 14.2; StataCorp, College Station, TX, USA).

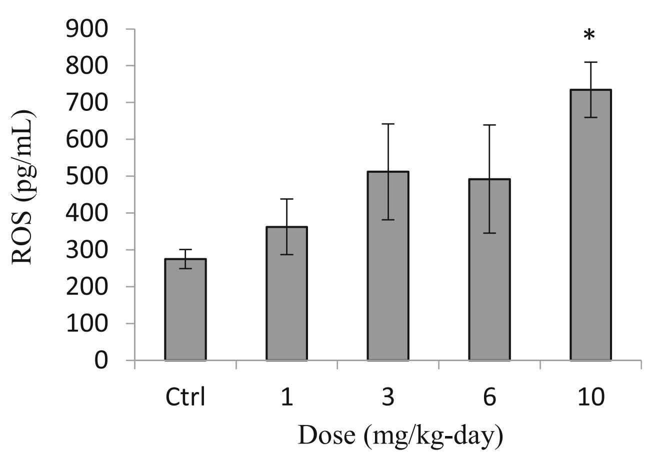


Fig. 2 The levels of ROS in the serum of mature male rats exposed orally to PS-NPs (0, 1, 3, 6, and 10 mg/kg-day) for 35 days. Data show mean values and associated standard deviation (SD) of 6 rats. Asterisks show significant differences ($p < 0.05$) from control value

Results and discussion

Reactive species and oxidative stress

Following exposure to PS-NPs, there observed a tangible dose-dependent increase of ROS levels in exposed rats (Fig. 2). As can be seen, there was a clear trend of separation between the control and PS-NPs exposure groups with approximately 167 % increase (p = 0.001, n^2 = 0.72) in ROS activity in the highest dose of 10 mg/kg-day. Exposure to smaller dosages caused an increase of ROS; however, the observed perturbations were not statistically appreciable ($p > 0.05$). Several macromolecules in the body including proteins, DNA, and lipids are susceptible to the oxidative damage caused by chemically reactive compounds containing oxygen (Apel and Hirt 2004). The cell/tissue sources of serum ROS in our study however is not clear; i.e., were there ROS elevation in tissues such as muscle and liver, or the blood cells. Impairment of mitochondrial respiration, the activation of NADPH oxidases or even inflammatory responses could also be involved, which are our current and future research interest. The literature already includes several reports of ROS generation and the subsequent toxic effects such as genetic, mitochondrial, and DNA damages in toxicity studies with engineered metal (oxide) nanoparticles (Khan et al. 2012; Shukla et al. 2011; Dawood et al. 2019). Two studies on 30-day exposure of male rats to TiO_2 and iron oxide NPs found significantly higher GSH level in the exposed group than in the controls (Kandeil et al. 2020; Mohammed et al. 2020). Another study suggested a significantly higher T3 level in zebrafish larvae exposed to ZnO nanoparticles. Similarly, elevated amounts of T3 were observed in zebrafish larvae on cobalt ferrite (CoFe2O4) nanoparticle treatment. A number of studies have also, in agreement with our findings, reported the induction of oxidative stress caused by exposure to small pieces (~20 μm) of polypropylene plastics (Hwang et al. 2019).

CAT and SOD activities

To obtain a broader picture, the treatment effects of PS-NPs are compared with a series of enzymes activities to quantify the level of oxidative stress (Fig. 3). The effect of PS-NPs on catalase activity is presented in Fig. 3a. Catalase activity was increased slightly following exposure experiments. The increase in CAT activity is possibly related to the defense mechanism against nanoparticles exposure. It is, also, supported by the increase in the ROS content as observed in Fig. 2. CAT plays a fundamental role in decomposition and reduction of H_2O_2 to water and oxygen and is responsible to directly scavenge reactive oxygen species. Although the difference was marginally significant (p = 0.07), it was possible to draw a meaningful conclusion relating to the impact of NPs exposure on CAT activity pointing toward the large effect size (n^2 = 0.72) obtained. However, with a small sample size, caution must be applied as the results of CAT analysis are unreliable based on the confidence interval results, demonstrating that the study needs to be replicated with a larger sample size or meta-analysis.

Although the literature on this topic has only just started to increase, but is somewhat contradictory with respect to the induction of oxidative stress and the antioxidant defense during exposure to microscopic plastic particles, some studies have reported altered oxidative stress biomarkers concentrations, for instance, increased activity of CAT following MPs exposure in clam *Scrobicularia plana* (Ribeiro et al. 2017), scleractinian coral *Pocillopora damicornis* (Tang et al. 2018), and African catfish (*Clarias gariepinus*) (Iheanacho and Odo 2020). This same general trend was reported by Revel et al. (2019) who pointed towards an appreciable increase in CAT activity in blue mussel *Mytilus* spp. exposed to a mixture of microplastics, but a lack of changes in ROS production. They explained this result by the sensitivity of exploited technique (flow cytometry) and/or the sampling time chosen for its measurement. However, polystyrene MPs in juvenile *Eriocheir sinensis* and larval zebrafish significantly alleviated the activity of CAT (Wan et al. 2019; Yu et al. 2018). In contrast, in a study involving *Mytilus galloprovincialis*, Sendra et al. (2019) reported a decreased percentage of ROS as a result of exposure to different sizes of PS-NPs. We would like to stress here that the variations in CAT activity in the literature may vary with species used and also are largely dependent on the particles concentration, size, and type used for exposure.

Perturbations in SOD activity noticed following exposure to PS-NPs did not reach statistical significance (p = 0.4) (Fig. 3b). In view of the ES (n^2 = 0.21), aiming at estimating the clinically important differences between groups, the activities of CAT and SOD ranged from 0.26 to 0.88 μmol/mL and 91 to 114 U/mL, respectively. SOD, the pioneering detoxifying enzyme catalyzes the breakdown of ROS-generating superoxide anion (O_2^-) and is a key component within the primary defense system against oxidative stress-induced damage. Contrary to expectations, this observation is not in line with the increase in systemic ROS level, since SOD is a sensitive enzyme to the ROS stress by superoxide. The large effect size with meaningful changes put forward that the generation of O_2^- may act a prime cause of oxidative impairment. Since μ_0 is inside the interval, it was not possible to draw a meaningful conclusion relating to the increased activity of SOD in the present study, and thus, this result should be again repeated with higher sample size or meta-analysis should be done. This result, however, could be assigned to the plausible inflammation and destabilization of lysosomal membranes, as suggested by Ribeiro et al. (2017). To the best of our knowledge, to date only one study has pointed out the toxic effects

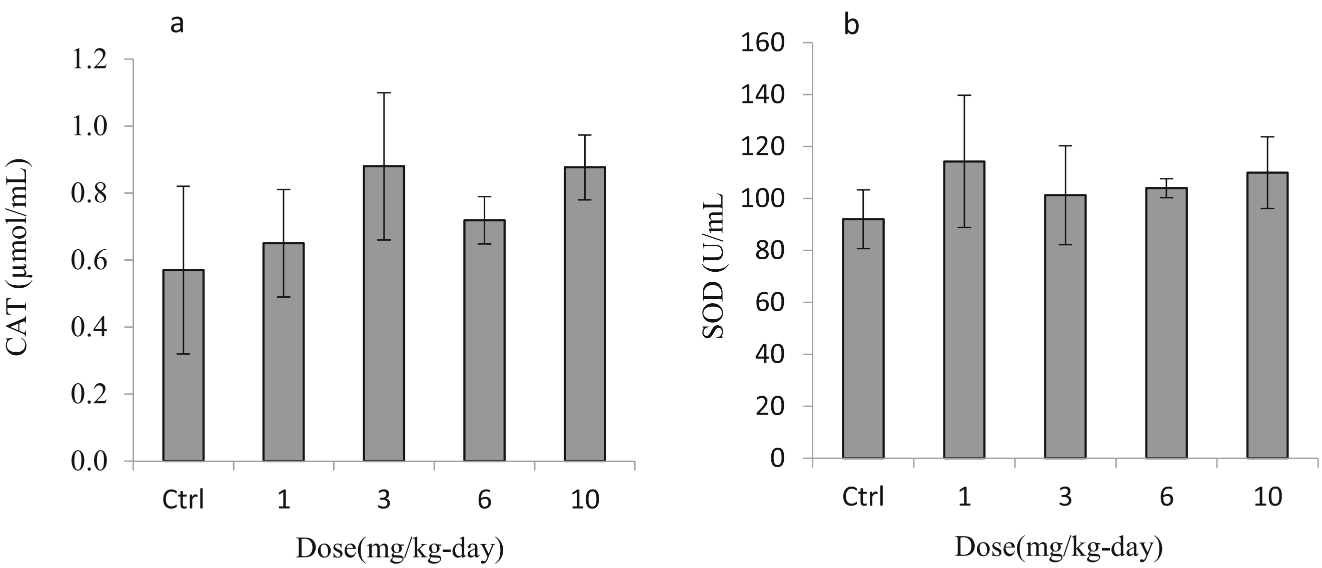


Fig. 3 The activities of **a** CAT and **b** SOD in the serum of mature male rats exposed orally to PS-NPs (0, 1, 3, 6, and 10 mg/kg-day) for 35 days. Data show mean values and associated standard deviation (SD) of 6 rats

of MPs on SOD activity in rodent species (Deng et al. 2017), and there is a paucity of literature on NPs. Compared to the negative control and tris (2-chloroethy) phosphate (TCEP) alone in Deng's study, the binary mixture of PS-MPs and TCEP induced the lowest activity in mice (*Mus musculus*). Rodent studies, however, have demonstrated alterations in SOD activity due to engineered nanoparticles exposure early in life or in adulthood (Abdel-Magied and Shedid 2019; Chen et al. 2020). Likewise, in the intestine of male marine medaka *(Oryzias melastigma*), chronic exposure to PS particles (10 μm) led to increased activity of SOD, while in females its activity was significantly decreased in ovaries (Hwang et al. 2019). Additionally, PS microbeads caused remarkable increase of SOD activity in zebrafish *Danio rerio* (Lu et al. 2016). In this context, Ribeiro et al. (2017) worked on the effects of clam exposure to PS-MPs on oxidative stress biomarkers. From the battery of biomarkers (SOD, CAT, GST, and GSH), only SOD was marked as the most prominent, and other parameters did not show clear patterns of change. Given the conflicting data reported, clarity on the perturbations in SOD activity from exposure to nano(micro)plastics awaits outcome of further studies.

GSH content

As shown in Fig. 4, the content of GSH declined markedly following exposure to PS-NPs ($p = 0.05, n^2 = 0.35$), though the significance lost at the highest dose. ROS accumulation is prevented by GSH, the reduced form of glutathione, that contributes in the detoxification of lipid peroxide and H_2O_2 peroxides into less-toxic hydroxyl compounds (Kandeil et al. 2020). GSH, however, can be inactivated by the sizeable ROS production, resulting in the overwhelming antioxidant defense system. If the observed positive associations between exposure to PS-NPs and ROS production and a concurrent significant decline in GSH content in the present study are causal, they could be explained by increased GPX activity. It has been proved that GPX detoxifies hydrogen peroxide or organic hydroxy peroxides through consumption of GSH. Glutathione can be conjugated in detoxification processes (Martinoia et al. 1993), which may also lead to its consumption and thus could be argued as another explanation for its depletion. Meanwhile, GSH is used as a cofactor or substrate for GST (Barata et al. 2005). Alternatively, associations might be explained by a combination of oxidative stress and enzymatic activities of antioxidant defense. The literature on associations between exposure to nano(micro)plastics and GSH content in aquatic organisms is not conclusive. According to

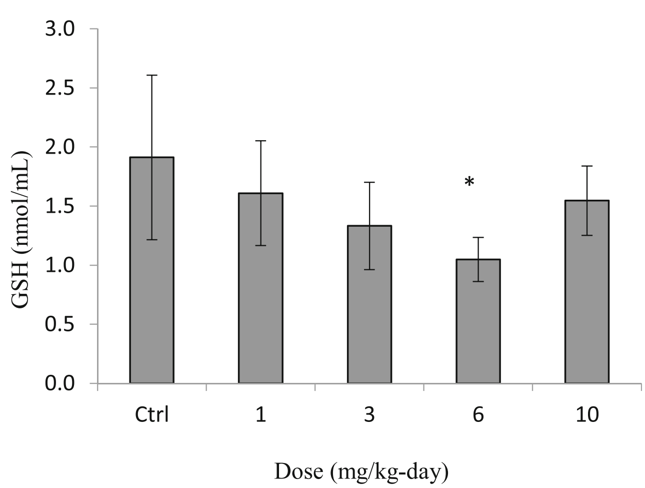


Fig. 4 The levels of GSH in the serum of mature male rats exposed orally to PS-NPs (0, 1, 3, 6, and 10 mg/kg-day) for 35 days. Data show mean values and associated standard deviation (SD) of 6 rats. Asterisks show significant differences ($p < 0.05$) from control value

Jeong et al. (2016), a toxicity study of fluorescently labeled polystyrene microbeads with different sizes (0.05, 0.5, and 6 μm) on Monogonont Rotifer (*Brachionus koreanus*) showed increased GSH content after 12 h in a size-dependent manner. Contrary to this study, however, declined GSH content has been reported in larval zebrafish (Wan et al. 2019) and juvenile *Eriocheir sinensis* (Yu et al. 2018) exposed to MPs. Overall, there are diverging results concerning GSH content and evidence on the relationship between GSH content and oxidative stress is not sufficient.

Acetylcholinesterase activity

Exposure to PS-NPs in our study resulted in AChE inhibition in all exposure levels compared to those of the normal rats, suggesting a neurotoxic effect of NPs; the highest dose (10 mg/kg-day) demonstrated a substantial and statistically significant partial effect (Fig. 5). An analysis of the data available in the literature already confirms the inhibition of AChE by ROS (Schülke et al. 2012). Given the involvement of acetylcholinesterase in neurological functions including some physiological (e.g., growth and reproduction) and behavioral (e.g., memory) processes and irrespective of the mechanism leading to AChE inhibition, the anticholinesterase poisoning of PS-NPs in rodent species provides an early warning sing of impacts at higher levels (e.g., organism and population). Nonetheless, the first evaluation of PS-NPs toxicity on the same subject demonstrated subtle and transient behavioral consequences in adult rats (Rafiee et al. 2018), denoting that neurotransmission might not be directly connected to the barely perceptible behavioral disorders. In support of our findings, Lin et al. (2019) observed declined AChE activity in *Daphnia magna* following exposure to various functionalized PS-NPs (PS-p-NH_2, PS-n-NH_2 and PS-COOH). They also reported a moderately positive but insignificant association between exposure to pristine plain PS-NPs and AChE activity.

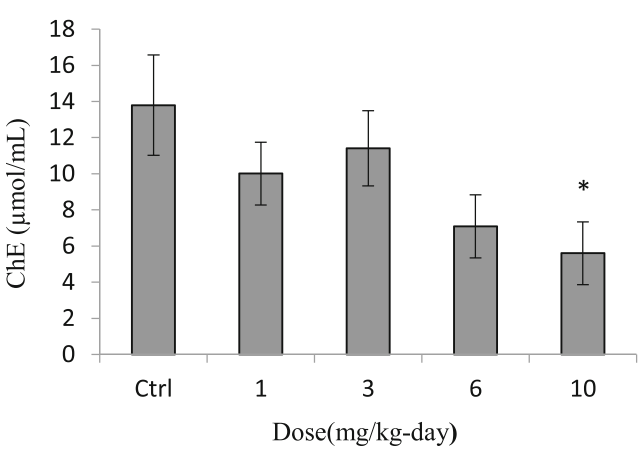


Fig. 5 The activities of AChE in the serum of mature male rats exposed orally to PS-NPs (0. 1, 3, 6, and 10 mg/kg-day) for 35 days. Data show mean values and associated standard deviation (SD) mean of 6 rats. Asterisks show significant differences ($p < 0.05$) from control value

A recent study by Ding et al. (2018) tried to assess the AChE activity of red tilapia (*Oreochromis niloticus*) exposed to 0.1 μm PS-MPs. They investigated different concentrations of PS-NPs (1, 10, and 100 μg/L) with adult fish for 14 consecutive days. There observed statistically significant reductions in AChE activity in fish brains exposed to PS particulates at all studied concentrations with maximum inhibition rate of 37.7%. Furthermore, although there was significant difference between the control and all exposed groups, no appreciable differences in AChE activity among samples treated with PS-NPs was perceived.

Likewise, in the study of Oliveira et al. (2013), the mixture of red polyethylene microspheres and pyrene induced the highest toxicity in juveniles of *Pomatoschistus microps* in terms of AChE activity inhibition, compared to MPs alone.

Biochemical responses

As shown in Fig. 6, there were differences in the levels of some biochemical parameters among different groups (see also Table 1). As can be seen, exposure to PS-NPs caused significant alterations in biochemical parameters associated with energy metabolism and stress response (glucose and cortisol) as well as liver function (GGT and ALP). Kidney disorders were also confirmed by the perturbation of creatinine and uric acid levels, as compared to the control group in a dose dependent manner. We attempted to establish a link between the biochemical alterations and the stress in exposed rats. Glucose and cortisol revealed statistically significant increment following exposure experiments. Cortisol plays a major role in the stress response and is widely used as a reliable biomarker in toxicology. This hormone is secreted from the teleost head kidney through the hypothalamo-pituitary-interrenal (HPI) axis (Pottinger 1998) to increase the energy availability during prolonged stress. This task is accomplished primarily through gluconeogenesis, resulting in higher levels of plasma glucose (Saravanan et al. 2011). The disruption of energy metabolism was indeed verified by the alteration of lactate where its levels showed significant ($p < 0.001$) increment with large effect size ($n^2 = 0.73$) following exposure to PS-NPs. Lactate is a normal product of anaerobic metabolism in the body, where pyruvate—the glycolysis product—is metabolized by LDH into lactate (Zhuang et al. 2020). In our study, a significant and meaningful increment in LDH activity was noticed in exposed rats as compared to the control group ($p = 0.001, n^2 = 0.7$), which could be attributed to necrosis of hepatocytes, heart failures, kidney dysfunction, renal cells necrosis, muscular dystrophy, anemia, and an increased glycolytic capacity (Chen et al. 2019). Cell membranes damage will indeed wield causal elevated levels of LDH in the blood (Banaee et al. 2014). A number of studies in cell cultures have,

Table 1 Turkey's multiple comparisons of serum biomarkers

PS NPs dose (mg/kg-day)	Total protein	Albumin (g/dL)	Globulin (g/dL)	Lactate (g/dL)	Lipase (u/L)	GGT (u/L)	Lactate dehydrogenase (u/L)	ALP (u/L)	Triglyceride (mg/dL)	Urea (mg/dL)	Glucose (mg/dL)
0	7^{a}	3.18^{a}	3.82^{a}	5.5^{a}	6.5^{a}	8.5^{a}	150^{a}	415.75^{a}	88.25^{a}	8_{a}	99.5^{a}
1	6.9^{a}	3.05^{ab}	3.84^{a}	6.5^{a}	12^{ab}	10.75^{a}	156^{a}	455.25^{ab}	89.35^{a}	38.5^{a}	153.5^{b}
3	5.13^{bc}	2.89^{bc}	2.22^{b}	10.75^{ab}	13^{bc}	14.75^{a}	167.75^{a}	484.25^{ab}	98^{ab}	44.75^{b}	155.5^{b}
6	4.95^{b}	2.94^{abc}	2^{b}	12.25^{b}	19^{cd}	15.25^{a}	165.25^{a}	492.5^{ab}	106.13^{b}	51.75^{c}	157.75^{b}
10	5.78^{c}	2.8^{c}	2.97^{c}	15.25^{b}	21.75^{d}	25.75^{b}	205^{b}	510.25^{d}	123^{c}	55.13^{c}	162.75^{b}

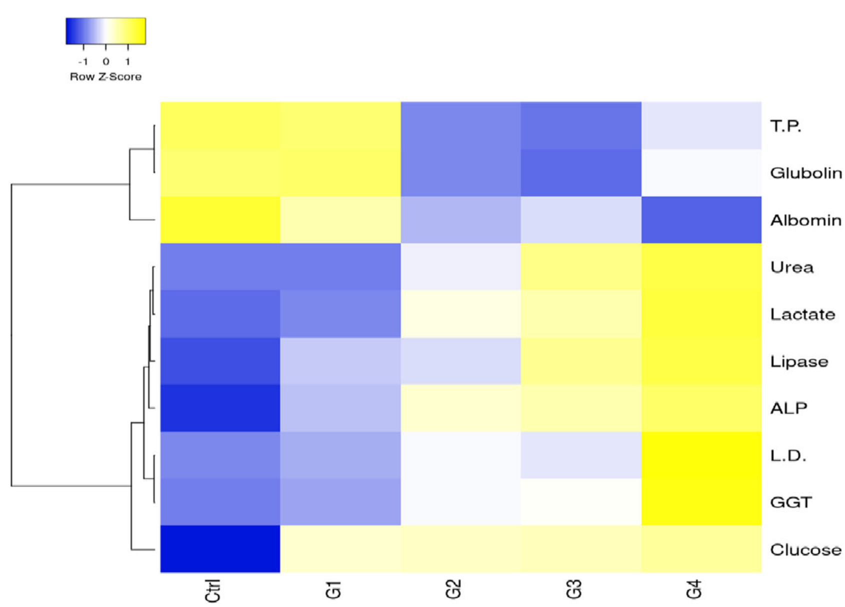
.Rats orally administered with PS NPs compared to the control group (6 animals in each group). Results are expressed as means. (*GGT*, gamma-glutamyl transpeptidase, *ALP*, alkaline phosphatase)

*Different letters for the significances demonstrate that there are significant differences among groups ($p < 0.05$)

in agreement with this hypothesis, show that nanoplastics rupture membranes (Grassi et al. 2020; Lu et al. 2016). In mice treated by PS-MPs, Deng et al. (2017) also observed an increasing trend in liver LDH activity with MPs exposure ($p < 0.05$). Similarly, in red tilapia, significantly higher LDH concentrations were observed after polystyrene MPs exposure (Ding et al. 2018). In contrast, Karami et al. (2016) found no association between activation of this enzyme and virgin microplastics exposure in African catfish. In a study on common carp (*Cyprinus carpio*) with and without exposure to microplastic particles, Haghi and Banaee (2017) did not notice a meaningful LDH activity. Similar results have also been reported in literature, demonstrating depletion in energy production as a consequence of MPs uptake both in toxicological and ecotoxicological studies (Green et al. 2016; Lu et al. 2016). Nevertheless, evidence regarding the effects of plastic particulates exposure and particularly NPs on energy metabolism is not conclusive.

Lactate is cleared from the blood, primarily by the liver and, to a lesser extent, by the kidneys and skeletal muscles. Hyperlactatemia and lactic acidosis, a serious and sometimes life-threatening condition, are reflected by imbalance between the systemic generation of lactate and its clearance, or a combination of both. However, all of the available literature have found that the lipase or lactate levels in living organisms did not significantly change when exposed to MPs (Ding et al. 2018; Karami et al. 2016). Although, ecotoxicological data on blood levels of lactate under nano(micro)plastics exposure are scarce; Karami et al. (2016) have reported no negative impact of pristine or phenanthrene-loaded polyethylene fragments on African catfish. The analysis of LDH provides a clearer picture whether—and to what extent—the energy metabolism was disrupted following exposure experiments. LDH activity has also been widely accepted as a good biomarker of alterations in energy pathways induced by xenobiotics (Al-Attas et al. 2015).

Fig. 6. Heatmap of biochemical parameters, which shows the different parameters in serum of rats. Each row represents a biochemical parameter, and each column represents a group. The color of each grid represents the relative concentration of the biochemical parameters in the corresponding group



Lipase, secreted from pancreas, performs essential roles in digestion, transport, and processing of dietary lipids (e.g., triglycerides, fats, oils) in most living organisms (Winkler et al. 1990). Statistically and clinically significant correlations observed between PS particles exposure in our study particularly at mid and highest dosages (3, 6, and 10 mg/kg-day) and lipase levels suggest that the pathways involved in cholesterol and triglycerides metabolism are correlated and that the damage to the pancreas cells could also impair serum levels of lipase in the body. The pathway could also possibly be related to the increase of cortisol, for which cholesterol is a precursor (Abdel-Magied and Shedid 2019).

Creatinine and uric acid are very sensitive biomarkers of kidney dysfunction serving as early warning signals for predicting adverse outcomes (Banaee 2013). The levels of these parameters quantified in the serum of rats were significantly higher with PS-NPs exposure at dosages of 3, 6, and 10 mg/kg-day; however, their levels were appreciably increased among those with highest exposure dosage. Uric acid is the end product of purine metabolism and a major antioxidant in humans. It can act as a prooxidant, especially at increased concentrations, and may therefore be an indicator of oxidative stress. However, uric acid may also perform a therapeutic role as an antioxidant (Strazzullo and Puig 2007). The observed positive associations between PS-NPs and a concurrent dose-dependent increase in creatinine and uric acid levels could be explained by dose-response kidney deterioration. Some studies in rats have reported increased creatinine concentrations as a consequence of kidney damage compared with control subjects (Rana et al. 2018). It is somewhat surprising that translocation of fluorescent PS-NPs from gastrointestinal tract to kidney has been previously reported in rats (Walczak et al. 2015), and also possibly in our work (Fig. 7). Increased levels of creatinine have also been reported in common carp exposed to sublethal concentrations of MPs and/or paraquat (Haghi and Banaee 2017). Serum uric acid to creatinine ratio may indeed be used as a marker in the pathogenesis of metabolic syndrome as well as to clarify the impairment of the kidney independently of potential confounders. As the increase of exposure dosage, the corresponding ratio increased. While cutoff values have not been defined, the calculated ratio in the present study paves the way to verify above viewpoints.

The increase in serum levels of urea, as shown in Table 1, provides further evidence for the toxicity of PS-NPs to the liver; as urea is the end product of protein catabolism, which is in good agreement with protein concentrations. In line with these results, Wright and Kelly (2017) revealed that MPs can accumulate and exert dose-dependent localized-particle toxicity by inducing inflammation and immune mechanisms in human.

At exposure dosages lower than 6 mg/kg-day, ALP activity increased slightly (Table 1). However, PS-NPs at 10 mg/kg-day remarkably increased the ALP activity. An ANOVA and the subsequent Tukey's multiple comparison analyses revealed that there were no appreciable differences in ALP activity among rats treated with PS-NPs at dosages ≤ 6 mg/kg-day; however, the activity was significantly increased by 10 mg/kg-day (Table 1). Moreover, based on the large effect size ($n^2 = 0.49$) and confidence interval, the effect of NPs on ALP activity is considerable and conclusive. As a cellular membrane enzyme, ALP catalyzes the dephosphorization of nucleotides, proteins, and alkaloids at an alkaline pH (Karami et al. 2016). This enzyme may be liberated into the plasma because of the degeneration and necrosis of hepatocytes following exposure to NPs (Iavicoli et al. 2012). The increased activity of ALP in exposed animals thus may be speculated by damage to the membrane of liver cells, bile ducts, intestine mucous lining, and the renal tract (Karami et al. 2016). In contrast to our findings, the impact of MPs at high dosages in common carp (at 2 mg/L) (Haghi &Banaee 2017) and red tilapia (at 200 μg/L) (Ding et al. 2018) on ALP activity has been studied, demonstrating decreased ALP activity; but the impact of nanoscale plastic particulates exposure particularly on mammalians is still unknown. Karami et al. (2016) also reported no magnification or diminution in plasma ALP activity of African catfish because of exposure to LDPE MP_S. By and large, these studies reflect some dearth in the digestive capabilities of exposed animals.

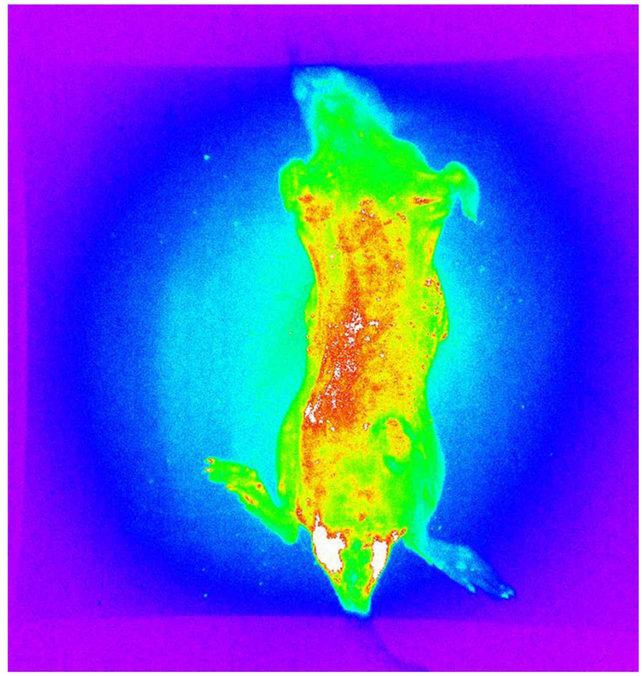


Fig 7 Image of fluorescently labeled polystyrene microplastic (10 mg/kg-day) accumulated in liver of rat

GGT activity reveals the same pattern, demonstrating a statistically and clinically significant increase in the highest exposure dosage ($p < 0.001$, $n^2 = 0.74$). The augmented activity of GGT is explained by its sensitivity to environmental insults (Mehrpak et al. 2015). Also, GGT is often the first hepatic enzyme to rise in the blood as a consequence of bile ducts obstruction, which take bile from the liver to the intestines. Accordingly, it is the most sensitive bioindicator in the body for detecting bile duct troubles, which is in line with visible accumulation of labeled PS-NPs in the liver (Fig. 7). While it could not discern exactly from this figure where NPs are and whether these are stagnant in the gut lumen or have penetrated cells or bioaccumulate in tissues/organs, we could see signals largely in the testes, thorax, and abdominal cavity providing evidence that the gut-blood barrier has been breached. However, GGT coupled with other tests provides a lot of useful advice in determining the plausible cause of elevated ALP levels. To test whether the increment in ALP has corresponded to bone tissue damage, we assessed the serum levels of GGT. As both GGT and ALP levels show increasing trends by exposure dosage, we speculate that the liver is damaged. If only ALP had increased, it would have indicated bone tissue damages. Therefore, GGT was used as a follow-up to an elevated ALP to help in determining the damaged liver or the bile ducts. This was in line with ROS (Fig. 2), where an increase in PS-NPs doses is associated with oxidative stress that is a strong inducer of ALP in various tissues (Torino et al. 2016). Perturbations in plasma GGT activity were also observed in common carp where its activity decreased when exposed to microplastic particles (Haghi &Banaee 2017). The 35-days exposure of rats to PS-NPs resulted in a remarkable and nonignorable dose-dependent decrease in the serum total protein, albumin, and globulin levels, while the ratio of albumin/globulin (A/G) significantly increased at 3 and 6 mg/kg-day exposed groups. The obvious reduction in serum total protein, albumin, and globulin levels could endanger efficient absorption of amino acids in the intestine (Abdel-Magied and Shedid 2019). The observed alterations might also be caused by impairments in protein digestion and absorption process and declined protein synthesis in the liver (Tang et al. 2016). Although firm evidence is lacking, however, the decrease in plasma proteins may have something to do with attachment to the particles surface and not free in the circulation. It could also result from impaired renal capacity especially in the wake of kidney impairment or even liver function. Nevertheless, the mechanism behind the declined serum proteins in this work remains unidentified.

Albumin and globulin decrement may also be assigned to the reduced protein synthesis in the liver of NPs-exposed animals. Decreased albumin levels may indeed result in osmoregulation defaults in blood (Karami et al. 2016). The observed dose-dependent decreasing pattern in globulin levels could be attributed to the reduced immunity in exposed animals (Ahmad et al. 2010; Egemen and Şeker 2003). The high ratio of A/G points towards the plausible liver and kidney tissues deterioration, which is in good agreement with the results of poor liver function. In line with our findings, exposure to MPs had induced alterations in the levels of total protein, globulin, and A/G ratio of African catfish, common carp (Karami et al. 2016; Haghi &Banaee 2017), and gilthead sea bream (*Sparus aurata L.*) (Beltrán et al. 2017).

The increased circulatory triglyceride levels of exposed animals in our study (Table 1) could also be assigned to liver lesions (Begriche et al. 2011; Karami et al. 2016) or plausible disturbance in its uptake in the intestinal tract (Reis et al. 2008). Some studies have reported altered triglycerides concentration in the blood (Cedervall et al. 2012, Haghi &Banaee 2017); while a lack of changes in serum levels of triglycerides has also been reported (Brandts et al. 2018). Triglycerides are needed to maintain the energy balance. In fact, the liver is the pioneering organ regulating the metabolism of triglycerides (Barnhart 1969).

In order to test whether the observed interruption in energy metabolism with PS-NPs exposure was correlated with oxidative stress, we conducted additional analyses where we noticed ROS content was positively correlated with glucose (r = 0.67, p = 0.002), triglycerides (r = 0.63, p = 0.004), lactate (r = 0.71, p = 0.001) as well as lactate dehydrogenase (r = 0.67, p = 0.002), demonstrating that elevated levels of these parameters did induce dysfunction and apoptosis in the body, possibly through the formation of reactive oxygen species. However, studies underlying the relationship between energy metabolism and oxidative stress following exposure to nanoplastics are still extremely scarce.

Conclusions

The present study demonstrated the toxicity of PS-NPs to male Wistar rats at both the organism and biochemical levels. Biochemical endpoints revealed that dietary exposure to nanosized PS can increase the oxidative status and cause vital damage (liver, kidney, and gut, among others). The present biochemical results revealed the increased activities of CAT, but not of SOD, with exposure to nanoscale plastic particulates. The content of GSH also decreased markedly following exposure to PS-NPs. Data from circulating concentrations showed declined AChE activity suggesting a possible neurotoxic action of PS-NPs. The pattern of associations noticed with AChE activity and biochemical responses in the present study point towards the possibility that neurobehavioral effects of nanoplastics or dysfunctions in energy metabolism, or both, may be the potential modes of action, possibly through stress response as well as liver function. Perturbation of creatinine and uric acid levels is also plausible biological explanations for the association with kidney dysfunction. Results point to the potential risks of nanoscale plastics exposure, their capability to induce physiological alterations and interfere in molecular pathways in mammalian species including human. However, with a small sample size, caution must be applied in the judgment of our findings, demonstrating that additional research is needed with a larger sample size.

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Data availability The datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate The local Ethics Committee of Ahvaz Jundishapur University of Medical Sciences approved the consent form.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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